Related Roadmaps

- ✓ DevOps Roadmap
- ✓ System Design Roadmap
- ✓ Shell/Bash Roadmap
- ✓ Cybersecurity Roadmap

Network Engineer

Find the detailed version of this roadmap
along with other similar roadmaps

roadmap.sh

Introduction

Basic Terminology

- Client
- Server
- Host
- Package
- Frame
- Bandwidth
- Latency
- Throughput
- Protocol
- Port
- Socket
- IP Address
- Mac Address
- ARP
- Transmission Media Types

Core Protocols

- HTTP / HTTPS
- SSH
- SSL / TLS
- NTP
- FTP / SFTP
- SNTP
- SMTP / IMAP
- DHCP

- How does the Internet Work?

What are Networks?

Network Types

- LAN
- WAN
- MAN
- WLAN
- PAN
- SAN
- VPN
- Cloud
- Client-Server Network
- Peer-to-Peer Network

Network Devices

- Routers
- Switches
- Hub
- Modems
- Access Points

OSI Model

7 Layers

- Physical
- Data Link
- Network
- Transport
- Session
- Presentation
- Application

TCP/IP Model

- Network Access
- Internet
- Transport
- Application

IP Addressing

- IPv4 vs IPv6
- Public vs Private Addresses
- IP vs MAC vs ARP

Subnetting

- Subnet Masks
- CIDR
- VLSM
- Supernetting

Routing

- Static vs Dynamic Routing
- Default Gateway

Routing Protocols

- BGP
- OSPF
- RIP
- EIGRP
- MPLS

Switching

- VLANs
- STP
- Link Aggregation
- MAC Address Tables

DNS

DNS Servers

- Cloudflare
- Google
- OpenDNS
- Quad9

Wireless Networking

- WiFi Standards
- Access Points & Controllers
- Bluetooth Basics
- Hotspot and Tethering
- Mobile Networks
- Wireless Security
  - WPA vs WPS

Load Balancer

- Round Robin
- Least Connections
- Failover

QoS (Quality of Service)

- Traffic shaping
- Packet prioritization

Network Security

Cybersecurity Roadmap

Firewalls

Firewall Types

- Packet Filtering
- Stateful Inspection
- Next-Generation
- Proxy
- Circuit Level Gateway
- Web Application

VPNs

- IPSec vs SSL VPN
- Site-to-Site vs Remote Access

Encryption Basics

Network Attacks

- DoS & DDoS

IDS / IPS

Zero Trust Architecture

Observability

Network Observabiity Tools

- Wireshark
- Nmap
- NetFlow / sFlow
- SNMP

Visit the following roadmaps to keep learning

DevOps | Backend | System Design